CV

Curriculum vitae of Kasemsit Teeyapan.

Contact Information

Name	Kasemsit Teeyapan
Professional Title	Assistant Professor, Ph.D.
Email	kasemsit.t@cmu.ac.th
Location	239 Huay Kaew Road, Suthep, Mueang, Chiang Mai, Chiang Mai 50200

Professional Summary

Assistant Professor at the Department of Computer Engineering, Faculty of Engineering, Chiang Mai University. Current research focuses on forensic odontology and computer vision-based reading of medical devices, alongside broader work in medical image analysis, computer vision, and machine learning.

Education

2012 - 2018
Chiang Mai, Thailand

Ph.D.

Chiang Mai University

Electrical Engineering

- Thesis: “Twin Hyper-ellipsoidal Support Vector Classifier”
- Advisor: Prof. Dr. Nipon Theera-Umpon

2007 - 2008
Atlanta, GA, USA

M.S.

Georgia Institute of Technology

Electrical and Computer Engineering

2001 - 2005
Chiang Mai, Thailand

B.Eng.

Chiang Mai University

Electrical Engineering (First-class Honors, Gold Medal)

Experience

2022 - PRESENT
Chiang Mai, Thailand

Assistant Professor

Department of Computer Engineering, Faculty of Engineering, Chiang Mai University

2018 - 2022
Chiang Mai, Thailand

Lecturer

Department of Computer Engineering, Faculty of Engineering, Chiang Mai University

2017 - 2017
Leuven, Belgium

Doctoral exchange student

Skeletal Biology and Engineering Research Center, KU Leuven Campus Group T

Worked with Veerle Bloemen on single-cell tracking and image processing.

2009 - 2011
Atlanta, GA, USA

Graduate Research Assistant

Humanoid Robotics Lab, Georgia Institute of Technology

Worked with Mike Stilman to develop Golem Krang, a two-wheeled dynamically-stable humanoid robot, focusing on modeling and control that exploit the robot's natural dynamics.

2005 - 2005
Ayutthaya, Thailand

Lens production engineer

Nikon (Thailand) Co., Ltd.

Supervised the manufacturing processes of interchangeable lenses.

Publications

International Journals

ISD — **Deep learning-based method for estimating age from periapical radiographs of upper incisors in a Thai population**
Praewpannarai Khruasarn, Kasemsit Teeyapan, Sangsom Prapayasatok, and Sakarat Nalampang
Imaging Science in Dentistry, Dec 2025

JAMS — **Application for acoustic assessment: A pilot study in Parkinson's patients**
Worapol Boonyaban, Piyawat Trevittaya, Nipon Theera-Umpon, Kasemsit Teeyapan, Chayasak Wantaneeyawong, and Anuwat Boonsong
Journal of Associated Medical Sciences, Oct 2024

IJERPH — **Fuzzy K-nearest neighbor based dental fluorosis classification using multi-prototype unsupervised possibilistic fuzzy clustering via cuckoo search algorithm**
Ritipong Wongkhuenkaew, Sansanee Auephanwiriyakul, Nipon Theera-Umpon, Kasemsit Teeyapan, and Uklid Yeesarapat
International Journal of Environmental Research and Public Health, 2023

JIFS — **A twin-hyperellipsoidal support vector classifier**
Kasemsit Teeyapan, Nipon Theera-Umpon, and Sansanee Auephanwiriyakul
Journal of Intelligent and Fuzzy Systems, Dec 2018

NCA — **Ellipsoidal support vector data description**
Kasemsit Teeyapan, Nipon Theera-Umpon, and Sansanee Auephanwiriyakul
Neural Computing and Applications, Dec 2017

International Conferences with Full Articles

KST — **A Comparative Study of Deep Learning Models for Cabbage Detection and Counting in Drone Imagery**
Panyapat Wongdee and Kasemsit Teeyapan
In 2025 17th International Conference on Knowledge and Smart Technology (KST), Feb 2025

CSBio — **Deep learning-based approach for corneal ulcer screening**
Kasemsit Teeyapan
In The 12th International Conference on Computational Systems-Biology and Bioinformatics (CSBio2021), Oct 2021

ICSEC — **Abnormality Detection in Musculoskeletal Radiographs using EfficientNets**
Kasemsit Teeyapan
In 2020 24th International Computer Science and Engineering Conference (ICSEC), Dec 2020

ICCSCE — **Application of support vector based methods for cervical cancer cell classification**
Kasemsit Teeyapan, Nipon Theera-Umpon, and Sansanee Auephanwiriyakul
In 2015 IEEE International Conference on Control System, Computing and Engineering (ICCSCE), Nov 2015

ICRA — **Robot limbo: Optimized planning and control for dynamically stable robots under vertical obstacles**
Kasemsit Teeyapan, Jiuguang Wang, Tobias Kunz, and Mike Stilman
In 2010 IEEE International Conference on Robotics and Automation, May 2010

Humanoids

Optimized control strategies for wheeled humanoids and mobile manipulators
Mike Stilman, Jiuguang Wang, Kasemsit Teeyapan, and Ray Marceau
In 2009 9th IEEE-RAS International Conference on Humanoid Robots.
Best paper award finalist , 2009

Short Peer-reviewed Conference Articles

Twin Support Vector based Classifier using Hyperellipsoids
Kasemsit Teeyapan, Sansanee Auephanwiriyakul, and Nipon Theera-Umpon
In International Conference on Green and Human Information Technology, Jan 2018

Twin Hyperellipsoidal Support Vector Classifier with Instance Reduction
Kasemsit Teeyapan, Sansanee Auephanwiriyakul, and Nipon Theera-Umpon
In International Conference on Information, System and Convergence Applications, Jan 2018

National Journal

การตรวจจับอาการง่วงนอนของผู้ขับขี่รถยนต์โดยใช้เทคนิคการประมวลผลภาพ
(Driver Drowsiness Detection using Image Processing Techniques)
ศุภวิชญ์ ละม่อม, นิพนธ์ ธีรอำพน, ศันสนีย์ เอื้อพันธ์วิริยะกุล, and เกษมสิทธิ์ ตียพันธ์
วารสารวิชาการเทคโนโลยีสุขภาพประเทศไทย (Thai Journal of Health Technology), 2022

Research Grants

- **Musculoskeletal abnormality detection for upper extremity** — The Murata Science Foundation (2019). *Principal Investigator.*
- **Machine Learning System Prototype for Digital Power Plant** — Electricity Generating Authority of Thailand (2018–2019). *Co-investigator; PI: Prof. Dr. Nipon Theera-Umpon.*
- **Development of Sound Source Localization and Speech Detection for Dinsow Mini Robot** — Thailand Research Fund (2018–2019). *Co-investigator; PI: Prof. Dr. Nipon Theera-Umpon.*

Awards

- 2025 **Winner, MED CHICKATHON 2025 Pitching Competition**
  Faculty of Medicine, Chiang Mai University
- 2017 **Erasmus+ / Erasmus Programme**
  European Commission
- 2007 **Royal Thai Government Scholarship**
  Ministry of Science and Technology, Thailand
- 2005 **Excellent Academic Award (Gold Medal)**
  Chiang Mai University
- 2004 **Best Academic Achievement Award in Electrical Engineering**
  The Engineering Institute of Thailand under H.M. the King's Patronage
- 2002 **Most Outstanding Freshman in Engineering**
  Prof. Dr. Tab Nilanidhi Foundation

Teaching

- **Undergraduate**
  - 259201 Computer Programming for Engineers (Python)
  - 261102 Computer Programming (C++)
  - 261306 Computer Engineering Probability and Statistics
  - 261405 Advanced Computer Engineering Technology
  - 269400 Advanced Information Systems and Network Technology
  - 261459 Deep Learning
  - 261499 Selected Topics in Computational Intelligence (Deep Learning)
  - 269382 Data Analytics for Non-IT Majors
  - 269481 Project Design in Information Technology
- **Graduate**
  - 261795 Selected Topics in Computational Intelligence (Deep Learning)

Service

- **Journal reviewer**
  - IEEE Transactions on Neural Networks and Learning Systems
  - IEEE Transactions on Artificial Intelligence
  - Engineering Applications of Artificial Intelligence
  - Displays
  - International Journal of Computer Information Systems and Industrial Management Applications (IJCISIM)
  - ECTI Transactions on Computer and Information Technology (ECTI-CIT)
  - Science & Technology Asia
  - Journal of Science and Technology, Mahasarakham University

- **Conference reviewer**
  - FUZZ-IEEE 2024
  - APSIPA ASC 2022
  - IIHMSP-FITAT 2020
  - FSDM 2018

- **Conference organizer**
  - Registration Co-Chair, APSIPA ASC 2022, Chiang Mai, Thailand (Nov 2022)
  - IEEE CIBCB 2016, Chiang Mai, Thailand (Oct 2016)

Certificates

- **Deep Learning Specialization** - *deeplearning.ai (Coursera)* (2018)
- **Japanese-Language Proficiency Test N2** - *Japan Foundation* (2018)
- **Japanese-Language Proficiency Test N3** - *Japan Foundation* (2015)

Languages

Thai : Native speaker

English : Professional working proficiency

Japanese : JLPT N2

Interests

Forensic odontology, Computer vision-based medical device reading: Medical image analysis, Computer vision, Machine learning, Deep learning, Data analytics, Robotics